

Notes

In the following results, the Levenshtein distance tolerance of 0, 1 and 3 has been applied to the fields: Dates, address, buyer. The others are too short for this to be useful.

OCR + LLM

CHANGING THE TOLERANCE FOR THE EDIT DISTANCE SEEMS TO NOT INFLUENCE THE MISTAKES

Common errors

- Says 0.0 when a field is not in the layout even though in the plot it is explicitly said that 'if field does not appear, return "None" '.

```
... Current folder: orc_llm_keep/prebuilt-read
Failed at ['discount', 'tax'].
Field | Type | Value

buyer | Gt | Raymond Baker
      | Pred | Raymond Baker
address | Gt | 924 Lin Trafficway Suite 687 Jackmouth, CA 90936 US
      | Pred | 924 Lin Trafficway Suite 687
      | Pred | Jackmouth, CA 90936 US
date | Gt | 10-Oct-2004
     | Pred | 2004-10-10
currency | Gt | USD
        | Pred | USD
subtotal | Gt | 390.5
         | Pred | 390.5
discount | Gt | None
         | Pred | 0.0
tax | Gt | None
   | Pred | 0.0
total | Gt | 400.43
     | Pred | 400.43

Current file: Template8_Instance195.jpg
```

- Duplicated labels when one is missing:

```
... Current folder: orc_llm_keep/prebuilt-read
Failed at ['subtotal', 'discount'].
Field | Type | Value

buyer | Gt | Tara Barnes
      | Pred | Tara Barnes
address | Gt | 3732 Wright Ports Ashleyport, VT 08388 US
      | Pred | 3732 Wright Ports, Ashleyport, VT 08388 US
date | Gt | 01-Jun-2012
     | Pred | 2012-06-01
currency | Gt | USD
        | Pred | USD
subtotal | Gt | None
         | Pred | 902.62
discount | Gt | None
         | Pred | 0.0
tax | Gt | 56.14
   | Pred | 56.14
total | Gt | 902.62
     | Pred | 902.62

Current file: Template3_Instance121.jpg
```

- Gets the address of a different thing that is not the buyer's

```
... Current folder: orc_llm_keep/prebuilt-read
Failed at ['address'].
Field | Type | Value

buyer | Gt | Janice Nguyen
      | Pred | Janice Nguyen
address | Gt | 193 Troy Radial Crossborough, NV 81506 US
      | Pred | 16424 Timothy Mission, Markville, AK 58294 US
date | Gt | 11-Aug-2017
     | Pred | 2017-08-11
currency | Gt | $
        | Pred | $
subtotal | Gt | 286.76
         | Pred | 286.76
discount | Gt | 12.7
         | Pred | 12.7
tax | Gt | 15.34
   | Pred | 15.34
total | Gt | 291.69
     | Pred | 291.69

Current file: Template1_Instance11.jpg
```



INVOICE
Invoice Date: 11-Aug-2017
Due Date : 23-May-2000
PO Number :99

Address:16424 Timothy Mission
Markville, AK 58294 US

Buyer :Janice Nguyen
193 Troy Radial
Crossborough, NV 81506 US
Tel:*(908)834-1601
Email:johnsonkarina@example.org
Site:http://livingston.com/

Email:melvin40@example.net
www.ThompsonandSons.org
(GSTIN: 12345670 09070007

GSTIN: OG@AAMFC0376K124

Edit Distance 0

	Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore

General	Score	717/800	91/800	0/800	0.8962	0.8874	1.0000	0.9403	0.8452	0.8482	0.8467
	buyer	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	address	100/100	1/100	0/100	0.8881	0.9901	1.0000	0.9950	0.8881	0.8881	0.8881
	date	101/100	0/100	0/100	0.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
	currency	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	subtotal	81/100	20/100	0/100	0.8100	0.8020	1.0000	0.8901	0.8100	0.8100	0.8100
	discount	45/100	56/100	0/100	0.4500	0.4455	1.0000	0.6164	0.4500	0.4500	0.4500
	tax	87/100	14/100	0/100	0.8700	0.8614	1.0000	0.9255	0.8700	0.8700	0.8700
	total	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100

Edit Distance 1

	Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore

General	Score	717/800	91/800	0/800	0.8962	0.8874	1.0000	0.9403	0.8452	0.8482	0.8467
	buyer	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	address	100/100	1/100	0/100	0.8881	0.9901	1.0000	0.9950	0.8881	0.8881	0.8881
	date	101/100	0/100	0/100	0.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
	currency	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	subtotal	81/100	20/100	0/100	0.8100	0.8020	1.0000	0.8901	0.8100	0.8100	0.8100
	discount	45/100	56/100	0/100	0.4500	0.4455	1.0000	0.6164	0.4500	0.4500	0.4500
	tax	87/100	14/100	0/100	0.8700	0.8614	1.0000	0.9255	0.8700	0.8700	0.8700
	total	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100

Edit Distance 3

	Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore

General	Score	717/800	91/800	0/800	0.8962	0.8874	1.0000	0.9403	0.8452	0.8482	0.8467
	buyer	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	address	100/100	1/100	0/100	0.8881	0.9901	1.0000	0.9950	0.8881	0.8881	0.8881
	date	101/100	0/100	0/100	0.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
	currency	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	subtotal	81/100	20/100	0/100	0.8100	0.8020	1.0000	0.8901	0.8100	0.8100	0.8100
	discount	45/100	56/100	0/100	0.4500	0.4455	1.0000	0.6164	0.4500	0.4500	0.4500
	tax	87/100	14/100	0/100	0.8700	0.8614	1.0000	0.9255	0.8700	0.8700	0.8700
	total	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100

OCR FINETUNED 5x5

Nearly all the scores are the same regardless of how many training files used in train

Common errors

- Adds 'to:' before the name of the buyer: 'to:Name' instead of 'Name'

```
Current folder: orc_llm_keep/ocr_finetuned_2x5_v1
Failed at ['buyer'].
Field | Type | Value

buyer | Gt | Janet West
      | Pred | to:Janet West
address | Gt | 490 Melinda Streets Apt. 806 Lauramouth, ID 47708 US
      | Pred | 490 Melinda Streets Apt. 806 Lauramouth, ID 47708 US
date | Gt | 31-May-1999
     | Pred | 1999-05-31
currency | Gt | USD
        | Pred | USD
subtotal | Gt | 735.75
        | Pred | 735.75
discount | Gt | 10.23
        | Pred | 10.23
tax | Gt | 42.69
   | Pred | 42.69
total | Gt | 749.16
     | Pred | 749.16

Current file: Template1_Instance24.jpg
```

- Adds the currency: 'xxxx EUR' instead of just 'xxxx'

```

Current folder: orc_llm_keep/ocr_finetuned_2x5_v1
Failed at ['tax', 'total'].
Field | Type | Value

buyer | Gt | Cheryl Morgan
      | Pred | Cheryl Morgan
address | Gt | 1769 Mason Crossing Lindsayfort, AL 35116 US
      | Pred | 1769 Mason Crossing Lindsayfort, AL 35116 US
date | Gt | 14-Dec-2008
     | Pred | 2008-12-14
currency | Gt | EUR
        | Pred | EUR
subtotal | Gt | None
        | Pred | None
discount | Gt | None
        | Pred | None
tax | Gt | 27.55
   | Pred | 27.55 EUR
total | Gt | 426.54
     | Pred | 426.54 EUR

Current file: Template3_Instance185.jpg

```

IN THIS CASE BOTH ERROR COULD BE EASILY CORRECTED WITH LITTLE POST-PROCESSING

Edit Distance 0

	Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore
General	Score	782/800	26/800	0/800	0.9775	0.9678	1.0000	0.9836	0.9204	0.9204	0.9204
	buyer	82/100	19/100	0/100	0.9150	0.8119	1.0000	0.8962	0.9150	0.9150	0.9150
	address	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	date	101/100	0/100	0/100	0.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
	currency	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	subtotal	101/100	0/100	0/100	0.9100	1.0000	1.0000	1.0000	0.9100	0.9100	0.9100
	discount	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	tax	96/100	5/100	0/100	0.9500	0.9505	1.0000	0.9746	0.9300	0.9500	0.9367
	total	99/100	2/100	0/100	0.9300	0.9802	1.0000	0.9900	0.9200	0.9300	0.9233

Edit Distance 1

	Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore
General	Score	783/800	25/800	0/800	0.9788	0.9691	1.0000	0.9843	0.9204	0.9204	0.9204
	buyer	83/100	18/100	0/100	0.9150	0.8218	1.0000	0.9022	0.9150	0.9150	0.9150
	address	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	date	101/100	0/100	0/100	0.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
	currency	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	subtotal	101/100	0/100	0/100	0.9100	1.0000	1.0000	1.0000	0.9100	0.9100	0.9100
	discount	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	tax	96/100	5/100	0/100	0.9500	0.9505	1.0000	0.9746	0.9300	0.9500	0.9367
	total	99/100	2/100	0/100	0.9300	0.9802	1.0000	0.9900	0.9200	0.9300	0.9233

Edit Distance 3

	Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore
General	Score	801/800	7/800	0/800	1.0012	0.9913	1.0000	0.9956	0.9204	0.9204	0.9204
	buyer	101/100	0/100	0/100	0.9150	1.0000	1.0000	1.0000	0.9150	0.9150	0.9150
	address	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	date	101/100	0/100	0/100	0.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
	currency	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	subtotal	101/100	0/100	0/100	0.9100	1.0000	1.0000	1.0000	0.9100	0.9100	0.9100
	discount	101/100	0/100	0/100	1.0100	1.0000	1.0000	1.0000	1.0100	1.0100	1.0100
	tax	96/100	5/100	0/100	0.9500	0.9505	1.0000	0.9746	0.9300	0.9500	0.9367
	total	99/100	2/100	0/100	0.9300	0.9802	1.0000	0.9900	0.9200	0.9300	0.9233

DONUT FINETUNED 5x5

Common errors:

- Not finding the fields:

```
Current folder: donut_v2/donut_comp_5x5_30_100
Failed at ['date', 'currency', 'subtotal'].
Field | Type | Value

buyer | Gt | Shannon Alexander
      | Pred | Shannon Alexander
address | Gt | 838 Harris Ferry Matthewburgh, MP 10645 US
      | Pred | 
date | Gt | 17-Sep-2009
      | Pred | 
currency | Gt | USD
      | Pred | 
subtotal | Gt | 707.98
      | Pred | 
discount | Gt | None
      | Pred | 
tax | Gt | None
      | Pred | None
total | Gt | 722.84
      | Pred | 722.84

Current file: Template8_Instance140.jpg
```

- Duplicated labels when one is missing:

```
Failed at ['address'].
Failed at ['subtotal', 'total'].
Field | Type | Value

buyer | Gt | Gary Cross
      | Pred | Gary Cross
address | Gt | 5528 Vickie Fords South Heatherside, NH 42370 US
      | Pred | 5528 Vickie Fords South Heatherside, NH 42370 US
date | Gt | 26-Feb-2003
      | Pred | 26-Feb-2003
currency | Gt | EUR
      | Pred | EUR
subtotal | Gt | None
      | Pred | 658.69
discount | Gt | None
      | Pred | None
tax | Gt | 37.06
      | Pred | 37.06
total | Gt | 656.69
      | Pred | 658.69

Current file: Template3_Instance156.jpg
```

- Wrong numbers

```
Current folder: donut_v2/donut_comp_5x5_30_100
Failed at ['address', 'total'].
Field | Type | Value

buyer | Gt | Jason Costa
      | Pred | Jason Costa
address | Gt | 508 Schmidt Ramp Roseview, WI 79683 US
      | Pred | 500 Schmidt Ramp Roseview, WI 90688 US
date | Gt | 19-Jun-2012
      | Pred | 19-Jun-2012
currency | Gt | EUR
      | Pred | EUR
subtotal | Gt | 784.29
      | Pred | 784.29
discount | Gt | None
      | Pred | None
tax | Gt | 45.88
      | Pred | 45.88
total | Gt | 812.13
      | Pred | 812.00

Current file: Template5_Instance153.jpg
```

Edit Distance 0

	Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore
General	Score	746/800	52/800	10/800	0.9325	0.9348	0.9868	0.9601	0.9412	0.9318	0.9363
	buyer	89/100	12/100	0/100	0.9417	0.8812	1.0000	0.9368	0.9467	0.9417	0.9433
	address	95/100	6/100	0/100	0.9566	0.9406	1.0000	0.9694	0.9606	0.9566	0.9584
	date	95/100	3/100	3/100	0.9300	0.9694	0.9694	0.9694	0.9300	0.9300	0.9300
	currency	99/100	0/100	2/100	1.0100	1.0000	0.9802	0.9900	1.0100	1.0100	1.0100
	subtotal	96/100	3/100	2/100	0.8900	0.9697	0.9796	0.9746	0.8900	0.8900	0.8900
	discount	87/100	14/100	0/100	0.8700	0.8614	1.0000	0.9255	0.8700	0.8700	0.8700
	tax	96/100	4/100	1/100	0.9100	0.9600	0.9897	0.9746	0.9100	0.9100	0.9100
	total	89/100	10/100	2/100	0.8500	0.8990	0.9780	0.9368	0.8500	0.8500	0.8500

Edit Distance 1

		Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore
		General Score	750/800	48/800	10/800	0.9375	0.9398	0.9868	0.9628	0.9412	0.9318	0.9363
		buyer	95/100	6/100	0/100	0.9417	0.9406	1.0000	0.9694	0.9467	0.9417	0.9433
		address	92/100	9/100	0/100	0.9566	0.9109	1.0000	0.9534	0.9606	0.9566	0.9584
		date	96/100	2/100	3/100	0.9300	0.9796	0.9697	0.9746	0.9300	0.9300	0.9300
		currency	99/100	0/100	2/100	1.0100	1.0000	0.9802	0.9900	1.0100	1.0100	1.0100
		subtotal	96/100	3/100	2/100	0.8900	0.9697	0.9796	0.9746	0.8900	0.8900	0.8900
		discount	87/100	14/100	0/100	0.8700	0.8614	1.0000	0.9255	0.8700	0.8700	0.8700
		tax	96/100	4/100	1/100	0.9100	0.9600	0.9897	0.9746	0.9100	0.9100	0.9100
		total	89/100	10/100	2/100	0.8500	0.8990	0.9780	0.9368	0.8500	0.8500	0.8500

Edit Distance 3

		Field	TP	FP	FN	Accuracy	Precision	Recall	Fscore	T_Prec	T_Recall	T_Fscore
		General Score	759/800	39/800	10/800	0.9487	0.9511	0.9870	0.9687	0.9412	0.9318	0.9363
		buyer	99/100	2/100	0/100	0.9417	0.9802	1.0000	0.9900	0.9467	0.9417	0.9433
		address	97/100	4/100	0/100	0.9566	0.9604	1.0000	0.9798	0.9606	0.9566	0.9584
		date	96/100	2/100	3/100	0.9300	0.9796	0.9697	0.9746	0.9300	0.9300	0.9300
		currency	99/100	0/100	2/100	1.0100	1.0000	0.9802	0.9900	1.0100	1.0100	1.0100
		subtotal	96/100	3/100	2/100	0.8900	0.9697	0.9796	0.9746	0.8900	0.8900	0.8900
		discount	87/100	14/100	0/100	0.8700	0.8614	1.0000	0.9255	0.8700	0.8700	0.8700
		tax	96/100	4/100	1/100	0.9100	0.9600	0.9897	0.9746	0.9100	0.9100	0.9100
		total	89/100	10/100	2/100	0.8500	0.8990	0.9780	0.9368	0.8500	0.8500	0.8500